Back to Exam

Answer Key

Instructor Use

ANSWER KEY · AP CALCULUS AB

Final Exam

Complete answer key with worked solutions for all 35 questions. Section A multiple choice answers are highlighted in green; Section B numerical answers and Section C free-response models are shown in full beneath each prompt.

29 MC Answers | 4 Numerical Solutions | 5 FRQ Worked Solutions

QUICK REFERENCE · MULTIPLE CHOICE ANSWERS

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
D	A	E	F	B	E	F	D	A
Q10	**Q11**	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**
F	D	B	C	E	B	F	D	A
Q19	**Q20**	**Q21**	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**
E	C	D	A	E	F	B	F	C
Q28	**Q29**							
D	E							

SECTION A · MULTIPLE CHOICE (1-29)

Question 1

FOUNDATIONS · LIMITS

Evaluate: $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$

A 0

B 3

C ∞

D 6 ✓ ANSWER

E 9

F Does not exist

Question 2

FOUNDATIONS · CONTINUITY

For $f(x) = (x^2 - 4)/(x - 2)$, what type of discontinuity exists at $x = 2$?

A Removable (hole) ✓ ANSWER

B Jump

C Infinite (asymptote)

D Oscillating

E f is continuous at $x = 2$

F Endpoint discontinuity

Question 3

FOUNDATIONS · LIMITS AT ∞

Evaluate: $\lim_{x \to \infty} \frac{3x^2 + 5x - 1}{6x^2 - 2x + 4}$

A 0

B 1/3

C 2

D 3

E 1/2 ✓ ANSWER

F ∞

Question 4

FOUNDATIONS · UNIT CIRCLE

Find: $\sin(7\pi/6)$

A 1/2

B √3/2

C −√3/2

D √2/2

E −√2/2

F −1/2 ✓ ANSWER

Question 5

THEOREMS · IVT

A function f is continuous on [1, 5] with $f(1) = -2$ and $f(5) = 8$. By the Intermediate Value Theorem, which value of k is GUARANTEED to have a corresponding c in (1, 5) where $f(c) = k$?

A k = −5

B k = 4 ✓ ANSWER

C k = 9

D k = 12

E k = −3

F None of the above

Question 6

THEOREMS · MVT

A function f is differentiable on [2, 8] with $f(2) = 4$ and $f(8) = 22$. By the Mean Value Theorem, there exists c in (2, 8) such that $f'(c) =$

A 1

B 2

C 6

D 9

E 3 ✓ ANSWER

F 18

Question 7

DIFFERENTIATION · POWER RULE

If $f(x) = 4x^3 - 2x^2 + 7x - 5$, find $f'(x)$.

A $4x^2 - 2x + 7$

B $12x^2 - 2x + 7$

C $12x^3 - 4x^2 + 7x$

D $x^3 - x^2 + 7$

E $4x^2 - 4x + 7$

F $12x^2 - 4x + 7$ ✓ ANSWER

Question 8

DIFFERENTIATION · PRODUCT RULE

If $f(x) = x^2 \cdot \sin(x)$, find $f'(x)$.

A $2x \cos(x)$

B $2x \cos(x) - x^2 \sin(x)$

C $2x \sin(x) - x^2 \cos(x)$

D $2x \sin(x) + x^2 \cos(x)$ ✓ ANSWER

E $x^2 \cos(x)$

F $2x^2 \cos(x)$

Question 9

DIFFERENTIATION · QUOTIENT RULE

If $f(x) = (3x + 1)/(x - 2)$, find $f'(x)$.

A $-7/(x - 2)^2$ ✓ ANSWER

B $3/(x - 2)^2$

C $(3x + 1)/(x - 2)^2$

D $-5/(x - 2)^2$

E $7/(x - 2)^2$

F $(6x - 7)/(x - 2)^2$

Question 10

DIFFERENTIATION · CHAIN RULE

If $f(x) = \sin(3x^2)$, find $f'(x)$.

A $\cos(3x^2)$

B $\cos(6x)$

C $3x \cos(3x^2)$

D $-6x \sin(3x^2)$

E $6x \sin(3x^2)$

F $6x \cos(3x^2)$ ✓ ANSWER

Question 11

DIFFERENTIATION · TRIG

What is the derivative of $f(x) = \tan(x)$?

A $\sec(x)$

B $\sec(x)\tan(x)$

C $-\csc^2(x)$

D $\sec^2(x)$ ✓ ANSWER

E $\cot(x)$

F $1/\cos(x)$

Question 12

DIFFERENTIATION · LOG

If $f(x) = \ln(5x^2 + 1)$, find $f'(x)$.

A $1/(5x^2 + 1)$

B $10x/(5x^2 + 1)$ ✓ ANSWER

C $(5x^2 + 1)/(10x)$

D $10x$

E $1/(10x)$

F $5x^2/(5x^2 + 1)$

Question 13

IMPLICIT DIFFERENTIATION

For the curve $x^2 + y^2 = 25$, find dy/dx.

A x/y

B $-y/x$

C $-x/y$ ✓ ANSWER

D y/x

E $-2x/y$

F $2y/x$

Question 14

IMPLICIT · PRODUCT TERM

For the curve $xy + y^2 = 6$, find dy/dx.

A $y/(x + 2y)$

B $y/(x - 2y)$

C $-y/(2y)$

D $(1 + 2y)/y$

E $-y/(x + 2y)$ ✓ ANSWER

F $1/(x + 2y)$

Question 15

APP. OF DERIVATIVES · EXTREMA

For $f(x) = x^3 - 3x^2 - 9x + 5$, at which x-value does a **local minimum** occur?

A $x = -3$

B $x = 3$ ✓ ANSWER

C $x = -1$

D $x = 0$

E $x = 1$

F $x = 5$

Question 16

APP. OF DERIVATIVES · L'HÔPITAL

Evaluate: $\lim_{x \to 0} \frac{\sin(2x)}{x}$

A 0

B $1/2$

C 1

D ∞

E Does not exist

F 2 ✓ ANSWER

Question 17

APP. OF DERIVATIVES · RELATED RATES

A spherical balloon's radius is increasing at **2 cm/s**. When **r = 5 cm**, how fast is the volume changing? [$V = (4/3)\pi r^3$]

A 25π cm³/s

B 50π cm³/s

C 100π cm³/s

D 200π cm³/s ✓ ANSWER

E 400π cm³/s

F $500\pi/3$ cm³/s

Question 18

APP. OF DERIVATIVES · CONCAVITY

For $f(x) = x^4 - 4x^3$, on which interval(s) is the graph **concave up**?

A $(-\infty, 0) \cup (2, \infty)$ ✓ ANSWER

B $(0, 2)$

C $(-\infty, 0)$ only

D $(2, \infty)$ only

E All real numbers

F $(0, \infty)$

Question 19

ANTIDERIVATIVES · +C RULE

Find: $\int(6x^2 - 4x + 3)\,dx$

A $2x^3 - 2x^2 + 3x$ (no +C)

B $12x - 4 + C$

C $6x^3 - 4x^2 + 3x + C$

D $2x^3 + 2x^2 + 3x + C$

E $2x^3 - 2x^2 + 3x + C$ ✓ ANSWER

F $2x^2 - 2x + 3 + C$

Question 20

ANTIDERIVATIVES · TRIG

Find: $\int\sec^2(x)\,dx$

A $\sec(x) + C$

B $\sec(x)\tan(x) + C$

C $\tan(x) + C$ ✓ ANSWER

D $-\cot(x) + C$

E $-\csc^2(x) + C$

F $\ln|\sec(x)| + C$

Question 21

U-SUBSTITUTION

Find: $\int 2x \cdot \cos(x^2)\,dx$

A $\cos(x^2) + C$

B $2\sin(x^2) + C$

C $-\sin(x^2) + C$

D $\sin(x^2) + C$ ✓ ANSWER

E $x^2 \sin(x^2) + C$

F $\sin(x) \cdot \cos(x) + C$

Question 22

U-SUB · DEFINITE

Evaluate: $\int_0^2 3x^2 \cdot e^{x^3}\, dx$

A $e^8 - 1$ ✓ ANSWER

B $e^2 - 1$

C e^8

D $e^8 - e$

E $8 - e$

F $e^3 - 1$

Question 23

RIEMANN SUM · LEFT

Approximate $\int_0^4 f(x)\, dx$ using a **LEFT Riemann sum** with n = 4 subintervals ($\Delta x = 1$) using the table:

x	0	1	2	3	4
f(x)	5	7	8	10	12

A 26

B 28

C 32

D 35

E 30 ✓ ANSWER

F 37

Question 24

RIEMANN · OVER/UNDER

For an **increasing** and **concave up** function, a **LEFT** Riemann sum produces an estimate that is:

A An overestimate

B Always exact

C An overestimate if Δx is small

D Depends on the function

E Sometimes over, sometimes under

F An underestimate ✓ ANSWER

Question 25

DEFINITE INTEGRALS · PROPERTIES

Given that $\int_0^3 f(x)\,dx = 7$ and $\int_3^5 f(x)\,dx = -2$, find $\int_0^5 f(x)\,dx$.

A −9

B 5 ✓ ANSWER

C −5

D 2

E 9

F 14

Question 26

DEFINITE INTEGRALS · GRAPHS

The graph of f consists of a **semicircle** of radius 3 above the x-axis from x = −3 to x = 3, plus a **triangle** below the x-axis with base 4 and height 2 from x = 3 to x = 7. Find $\int_{-3}^{7} f(x)\,dx$.

A $9\pi/2 + 4$

B $9\pi - 4$

C $9\pi/2$

D $4 - 9\pi/2$

E $9\pi + 4$

F $9\pi/2 - 4$ ✓ ANSWER

Question 27

FTC PART 1 · CHAIN RULE

If $g(x) = \int_{0}^{x^2} \sin(t)\,dt$, find $g'(x)$.

A $\sin(x^2)$

B $2x\cos(x^2)$

C $2x\sin(x^2)$ ✓ ANSWER

D $\cos(x^2) - 1$

E $\sin(x^2) - \sin(0)$

F $x^2\cos(x)$

Question 28

APP. OF INTEGRALS · AREA

Find the area of the region bounded by $y = 2x$ and $y = x^2$ from $x = 0$ to $x = 2$.

A $2/3$

B 1

C 8/3

D 4/3 ✓ ANSWER

E 2

F 4

Question 29

PHYSICS · SHM

A particle moves along the x-axis with position $x(t) = 4\cos(2t)$. What is its velocity at $t = \pi/4$?

A −4

B 0

C 4

D 8

E −8 ✓ ANSWER

F 8 sin(π/2)

SECTION B · MULTI-PART NUMERICAL (30)

Question 30

MULTI-PART · PARTICLE MOTION

A particle moves along the x-axis with velocity $v(t) = 3t^2 - 12t + 9$ for $0 \le t \le 5$.

Answer all four parts. Enter exact numerical values (no units).

PART (A)

The smaller value of t at which the particle is at rest:

1

PART (B)

The larger value of t at which the particle is at rest:

3

PART (C)

The total **displacement** from $t = 0$ to $t = 5$: $\int_0^5 v(t)\,dt$

20

PART (D)

The **acceleration** at $t = 2$: $a(2)$

0

SECTION C · FREE RESPONSE (31-35)

Question 31

FRQ · RELATED RATES

A 13-foot ladder leans against a vertical wall. The bottom of the ladder is sliding away from the wall at a rate of **2 ft/s**.

(a) How fast is the top of the ladder sliding down the wall when the bottom is **5 ft** from the wall?

(b) Justify why your answer to (a) must be negative, in context.

Question 32

FRQ · MVT JUSTIFICATION

A function f is continuous on **[2, 10]** and differentiable on **(2, 10)**, with the values:

T	2	4	7	10
f(t)	5	9	14	21

(a) Use the Mean Value Theorem to justify why there must exist a value **c** in **(2, 10)** such that **f'(c) = 2**.

(b) Use a left Riemann sum with the 3 subintervals shown to approximate $\int_2^{10} f(t)\,dt$.

Question 33

FRQ · AREA & VOLUME

Let R be the region in the first quadrant bounded by **y = √x** and **y = x/2**.

(a) Find the x-coordinates of the points where the two curves intersect.

(b) Find the area of region R.

(c) Find the volume of the solid generated when R is rotated about the x-axis.

Question 34

FRQ · GRAPH OF F'

The graph of **f'(x)**, the derivative of f, on **[0, 8]** consists of two pieces:

- A **semicircle** of radius 2 above the x-axis from x = 0 to x = 4
- A **line segment** from (4, 0) to (8, -4)

It is given that **f(0) = 5**.

(a) Find all x-values where f has a local maximum on (0, 8). Justify.

(b) Find **f(4)**.

(c) On what interval(s) is f concave down on $(0, 8)$? Justify.

Question 35

FRQ · TABLE & RIEMANN

The velocity of a car (in ft/s) is recorded at 5-second intervals:

T (SEC)	0	5	10	15	20
v(t)	30	45	50	55	40

(a) Use a midpoint Riemann sum with **2** equal subintervals to approximate $\int_0^{20} v(t)\,dt$.

(b) Interpret $\int_0^{20} v(t)\,dt$ in context, with units.

(c) Use a trapezoidal sum with the **4** subintervals shown to approximate $\int_0^{20} v(t)\,dt$.